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TAIWAN CCL/PCB

Much more favorable CCL/PCB/Substrate pricing outlook with lengthening lead times suggesting better earnings outlook; Buy on EMC/TUC/GCE/ZDT

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We see multiple new catalysts for CCL/PCB/second-tier ABF substrate suppliers, including better pricing outlook/ faster product mix improvement (thanks to AI adoption) / longer production lead times (all CCL and PCB, substrates are seeing 20+ weeks of order lead times now for the most shortage spec products), and expect EMC/TUC/GCE/ZDT to enjoy a much better earnings outlook in the long term. For ZDT specifically, we believe the company will start shipping the new Rubin computing tray, where we expect it to enjoy 10%+ market share, and will start working on Google PCIE PCB board; moreover, we expect ZDT to be one of the key suppliers for Google TPU 8 from 2H26, implying a much better ABF substrate ASP/GM outlook.

As a result, we raise our 12-m TP for **EMC from NT$3,200 to NT$3,500**, **TUC from NT$900 to NT$1,015**, **GCE from NT$1,060 to NT$1,210**, and **ZDT from NT$228 to NT$260**. Maintain Buy on all 4 CCL/PCB/Substrate names.

CCL industry update: Stronger than expected pricing outlook, with lengthening lead time

For CCL, in April, we estimate the pricing for low-end/high-end CCL pricing will be up 30%/15% QoQ the highest, driven mainly by the imbalanced supply demand conditions now, while we note low-end CCL product lead time is at least 20 weeks+ now and the high-end CCL product lead time for LTA/MOU projects is only at 4 weeks, but can be as high as 12+ weeks for non LTA/MOU projects. The much stronger than expected demand from the high-end industry continues to crowd out the resource for mid to low end CCL demand, suggesting a faster than expected product mix improvement speed. Also, our checks with EMC and TUC last week (see here) suggests another round of pricing hikes is highly possible to happen in 2H26, which would be another catalyst for EMC and TUC's revenue/profitability and earnings in coming quarters.

Moreover, we see potential opportunity for the Taiwan high-speed CCL suppliers to enter the substrate CCL market from 2H26, given the fact of T-glass supply constraints (see here), which should not only imply a much higher TAM for Taiwan CCL suppliers, but also imply a better product mix and GM outlook in long term (substrate CCL GM can be as high as 40%+ vs. EMC/TUC GM at 22-30% now).

On the other hand, we expect the demand on AI CCL to continue to go up in coming

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years (see here), which is leading TUC, while both of the companies mentioned the only reason for them to expand so much capacity in 2027 is driven by customers' product demand/outlook.

For EMC specifically, we estimate the company will start regaining market share back in the NVDA supply chain (mainly HGX UBB and NVL switch tray from Rubin generation + EMC will be the sole supplier for CPX/LPU/midplane) driven by the technology migration (we believe HGX UBB, CPX, LPU will be based on M9 grade CCL, and NVL switch tray + midplane will be based on either M8.5 grade CCL or M9 grade CCL,based on our supply chain check and the new product performance requirements), which should also drive EMC's overall CCL ASP in coming quarters.

Overall speaking, we believe CCL suppliers, especially the high-end CCL suppliers, will continue to enjoy solid growth in earnings, driven by the better pricing outlook

PCB industry update: Pricing hike to offset negative impact from higher CCL cost, while the demand is accelerating, benefiting tier-2 suppliers

For PCB, we believe the pricing hike in 1H26 (~10% across all sectors, excluding consumer electronics PCB) will fully offset the higher CCL cost, suggesting a much better than our previous expected GM estimate. Moreover, we continue to expect a shortage of high-end PCB capacity, leading GCE to continue to expand more capacity in next few years to better support customers demand, due mainly to (1) the increasing usecase on PCB in the AI server industry (board to board, tray to tray connectivity will also use PCB instead of datacom cable in new generation AI servers), (2) the increasing PCB size and layer count design (we believe both layer count and PCB size will continue to go up by 20-30% YoY every year, while backplane layer count can be 78+ layers vs. 24-28 layers in AI server industry now), and (3) the decreasing production yield rate (from 80-90% now, to 75-85% in 2026, which could further go down to 60-70% in 2027 based on our estimate).

Moreover, considering the increasing demand on high-end PCB and the slow capacity expansion in the high-end PCB suppliers side, we expect second tier players, who have more proactive expansion plans like ZDT, to start entering the high-end AI PCB industry, and to continue to enjoy a better product mix and ASP migration, suggesting a better earnings outlook. For example, we expect ZDT to enter than Rubin computing board industry from 2H26, with it also being the AWS Trainium OAM supplier, and we note it is working on some qualification with Google/META/MSFT etc. companies.

As a result, we are positive on the high quality high-end PCB suppliers' (including GCE and ZDT) earnings outlook, and they should continue to deliver better ASP, shipment, GM outlook in coming quarters.

ABF substrate industry update: ZDT to enter the high-end IC accelerator substrate industry

Based on our ZDT's earnings call (see here), the company expects its ABF substrate capacity to be fully utilized in 2H26, encouraged by the customers' new AI projects, and we believe it is likely to be the ABF substrate supplier for TPU v8 considering it's the only new AI project in 2H26 + ZDT has been Mediatek's (2454.TW; covered by Bruce Lu; Buy) long term partner in BT substrate; we believe such a project would be ZDT's first US high-end ABF substrate project (dollar content per substrate could be at least

US$50-100) with decent volume (we would expect the total shipment of ZDT's TPU substrate tp be at least 1 million from 2H26-1H27).

We believe ZDT's positive comments on the high utilization rate and new AI project echoes our view on the ABF substrate shortage outlook (see here); while the company used to be a tier 3 ABF substrate supplier, it stated that it can still fill all of its idle capacity + get new AI project wins, implying the tier 1 +2 players are all fully utilized, giving ZDT the opportunity to work with high-end customers and improve its production technology. We believe the ABF substrate shortage trend (which we expect will continue in at least 2026-28) will lead to solid profitability for ZDT (ABF GM/OPM can be as high as 50-60%/35-45% during upcycle vs. ZDT's ABF substrate was loss making in OP level in 2025) in coming quarters/years and drive the overall company's earnings.

Overall, we expect a good market share outlook for ZDT in the high-end AI server substrate industry, which should lead to a much better ASP and profitability. Together with the company's proactive capacity expansion plan on ABF substrate (we believe ABF capacity will double in 3 years), we expect ZDT's ABF substrate business to contribute 15-20% OPI in 2027-28 vs. ~5% in 2026 vs. loss making in 2025.

Earnings revision

EMC

We revise up our 2026/27/28E EPS estimate by 1/2/2% to factor in the much better than expected mid to low end CCL pricing outlook (~30% of total revenue), and revise up our revenue estimate by 1/1/1%, with better GM outlook (revise up 2026/27/28E GM by <0.1/0.2/0.2ppt).

Exhibit 1: EMC earnings revision table

EMC P&L (NT$ mn)	2026 New	2026 Old	Diff.	2027 New	2027 Old	Diff.	2028 New	2028 Old	Diff.
Sales	148,826	147,698	1%	273,041	269,714	1%	407,986	402,869	1%
Gross Profit	47,799	47,390	1%	94,243	92,600	2%	144,412	141,625	2%
EBIT	35,828	35,510	1%	75,244	73,833	2%	118,731	116,266	2%
Net Income	27,224	26,985	1%	56,433	55,375	2%	89,048	87,200	2%
EPS (NT$)	**76.13**	**75.47**	**1%**	**157.82**	**154.86**	**2%**	**249.03**	**243.86**	**2%**
Ratio analysis									
Gross margin	32.1%	32.1%	0.0pp	34.5%	34.3%	0.2pp	35.4%	35.2%	0.2pp
EBIT margin	24.1%	24.0%	0.0pp	27.6%	27.4%	0.2pp	29.1%	28.9%	0.2pp
Net margin	18.3%	18.3%	0.0pp	20.7%	20.5%	0.1pp	21.8%	21.6%	0.2pp

Source: Company data, Goldman Sachs Global Investment Research

TUC

We revise up our 2026/27/28E EPS estimate by 2/3/2% to factor in the much better than expected mid to low end CCL pricing outlook (~50% of total revenue), and revise up our revenue estimate by 1/2/2%, with better GM outlook (revise up 2026/27/28E GM by 0.1/0.3/0.1ppt).

Exhibit 2: TUC earnings revision table

TUC P&L (NT$ mn)	2026E New	2026E Old	Diff.	2027E New	2027E Old	Diff.	2028E New	2028E Old	Diff.
Sales	51,741	51,065	1%	85,009	83,263	2%	130,739	128,102	2%
Gross Profit	14,293	14,080	2%	26,364	25,548	3%	42,408	41,451	2%
EBIT	10,697	10,530	2%	21,495	20,780	3%	35,403	34,587	2%
Net Income	8,167	8,041	2%	16,121	15,585	3%	26,499	25,887	2%
EPS (NT$)	**28.35**	**27.91**	**2%**	**55.96**	**54.10**	**3%**	**91.98**	**89.86**	**2%**
Ratio analysis									
Gross margin	27.6%	27.6%	0.1pp	31.0%	30.7%	0.3pp	32.4%	32.4%	0.1pp
EBIT margin	20.7%	20.6%	0.1pp	25.3%	25.0%	0.3pp	27.1%	27.0%	0.1pp
Net margin	15.8%	15.7%	0.0pp	19.0%	18.7%	0.2pp	20.3%	20.2%	0.1pp

Source: Company data, Goldman Sachs Global Investment Research

GCE

We revise up our 2026/27/28E EPS estimate by 1/2/2% to factor in the better than expected high-end AI PCB industry pricing outlook (revise up 2026/27/28E revenue estimate by <1%/1%/1%), while the increasing pricing outlook will also suggest a faster product mix improvement progress (revise up 2026/27/28E GM by 0.2/0.2/0.3ppt).

Exhibit 3: GCE earnings revision table

GCE P&L (NT$ mn)	2026E New	2026E Old	Diff.	2027E New	2027E Old	Diff.	2028E New	2028E Old	Diff.
Sales	100,933	100,704	0%	159,195	157,850	1%	191,681	190,067	1%
Gross Profit	37,082	36,819	1%	62,715	61,797	1%	77,001	75,728	2%
EBIT	28,661	28,413	1%	49,899	49,090	2%	61,671	60,528	2%
Net Income	19,243	19,076	1%	34,190	33,629	2%	42,618	41,821	2%
EPS (NT$)	**38.49**	**38.15**	**1%**	**68.38**	**67.26**	**2%**	**85.24**	**83.64**	**2%**
Ratio analysis									
Gross margin	36.7%	36.6%	0.2pp	39.4%	39.1%	0.2pp	40.2%	39.8%	0.3pp
EBIT margin	28.4%	28.2%	0.2pp	31.3%	31.1%	0.2pp	32.2%	31.8%	0.3pp
Net margin	19.1%	18.9%	0.1pp	21.5%	21.3%	0.2pp	22.2%	22.0%	0.2pp

Source: Company data, Goldman Sachs Global Investment Research

ZDT

We revise down ZDT's 2026E EPS estimate by 5% to factor in the much stronger than expected CCL pricing outlook, which would dilute ZDT's 2026 GM by 0.5ppt. For 2027/28, considering the company's proactive capacity expansion plans in Thailand and mainland China to better compete with other AI PCB suppliers, we expect the company's AI revenue exposure will reach 10%/28%/35%+ in 2026/27/28E, leading us to revise up the 2027/28E revenue estimate by 4/5%. Moreover, considering the better ABF and BT pricing outlook, we revise up our ZDT 2027/28E GM by 0.7/0.8ppt, leading to a 9/12% higher EPS estimate.

Exhibit 4: ZDT earnings revision table

ZD Tech P&L (NT$ mn)	2026E New	2026E Old	Diff.	2027E New	2027E Old	Diff.	2028E New	2028E Old	Diff.
Sales	221,241	220,616	0%	287,676	277,306	4%	362,926	345,114	5%
Gross Profit	46,979	47,841	-2%	68,657	64,303	7%	91,187	83,807	9%
EBIT	21,009	22,167	-5%	38,239	34,973	9%	53,244	47,285	13%
Net Income	13,348	14,042	-5%	23,623	21,674	9%	32,575	29,023	12%
EPS (NT$)	**13.34**	**14.04**	**-5%**	**23.61**	**21.67**	**9%**	**32.56**	**29.01**	**12%**
Ratio analysis									
Gross margin	21.2%	21.7%	-0.5pp	23.9%	23.2%	0.7pp	25.1%	24.3%	0.8pp
EBIT margin	9.5%	10.0%	-0.6pp	13.3%	12.6%	0.7pp	14.7%	13.7%	1.0pp
Net margin	6.0%	6.4%	-0.3pp	8.2%	7.8%	0.4pp	9.0%	8.4%	0.6pp

Source: Company data, Goldman Sachs Global Investment Research

Valuation & TP revision

EMC

We revise up EMC's 12m TP from NT$3,200 to NT$3,500, to factor in our more positive view on the target P/E multiple (from 25x 4Q26-3Q27E EPS previously to 27x 4Q26-3Q27E EPS now, which is in-line with the AI tech component leader suppliers' peak multiple in he past 3 years), to better reflect the solid AI demand outlook into long term and the much more favorable than expected pricing outlook in coming quarters/years, and factor in our earnings revision.

Exhibit 5: EMC P&L table

NT$mn	1Q25	2Q25	3Q25	4Q25	1Q26E	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026E	2027E	2028E
Revenue	21,680	22,508	25,146	24,927	32,245	35,484	39,537	41,561	48,682	62,966	75,900	85,493	64,377	94,261	148,826	273,041	407,986
Gross profit	6,590	6,829	7,576	7,124	9,639	11,353	12,892	13,916	16,466	21,698	26,205	29,873	17,970	28,120	47,799	94,243	144,412
Operating expense	(2,051)	(2,186)	(2,586)	(2,190)	(2,580)	(2,945)	(3,163)	(3,283)	(3,797)	(4,408)	(5,237)	(5,557)	(5,818)	(9,012)	(11,971)	(18,999)	(25,681)
Operating income	4,540	4,644	4,990	4,935	7,059	8,407	9,729	10,633	12,669	17,291	20,968	24,316	12,152	19,108	35,828	75,244	118,731
Pretax income	4,667	4,467	5,122	4,620	7,059	8,407	9,729	10,633	12,669	17,291	20,968	24,316	12,133	18,876	35,828	75,244	118,731
Taxes expense	(1,200)	(989)	(1,157)	(884)	(1,412)	(2,102)	(2,432)	(2,658)	(3,167)	(4,323)	(5,242)	(6,079)	(2,564)	(4,231)	(8,604)	(18,811)	(29,683)
Net income	3,469	3,478	3,965	3,737	5,647	6,306	7,297	7,975	9,502	12,968	15,726	18,237	9,578	14,649	27,224	56,433	89,048
EPS, NT$	**10.01**	**10.02**	**11.19**	**10.45**	**15.79**	**17.63**	**20.41**	**22.30**	**26.57**	**36.27**	**43.98**	**51.00**	**27.81**	**41.67**	**76.13**	**157.82**	**249.03**
Ratio analysis and assumption																	
As % of sales																	
Gross margin	30.4%	30.3%	30.1%	28.6%	29.9%	32.0%	32.6%	33.5%	33.8%	34.5%	34.5%	34.9%	27.9%	29.8%	32.1%	34.5%	35.4%
Operating expense ratio	9.5%	9.7%	10.3%	8.8%	8.0%	8.3%	8.0%	7.9%	7.8%	7.0%	6.9%	6.5%	9.0%	9.6%	8.0%	7.0%	6.3%
Operating margin	20.9%	20.6%	19.8%	19.8%	21.9%	23.7%	24.6%	25.6%	26.0%	27.5%	27.6%	28.4%	18.9%	20.3%	24.1%	27.6%	29.1%
Net margin	16.0%	15.5%	15.8%	15.0%	17.5%	17.8%	18.5%	19.2%	19.5%	20.6%	20.7%	21.3%	14.9%	15.5%	18.3%	20.7%	21.8%
QoQ growth (%)																	
Revenue	16.8%	3.8%	11.7%	-0.9%	29.4%	10.0%	11.4%	5.1%	17.1%	29.3%	20.5%	12.6%	-	-	-	-	-
Gross profit	24.8%	3.6%	10.9%	-6.0%	35.3%	17.8%	13.6%	7.9%	18.3%	31.8%	20.8%	14.0%	-	-	-	-	-
Operating income	28.8%	2.3%	7.5%	-1.1%	43.1%	19.1%	15.7%	9.3%	19.2%	36.5%	21.3%	16.0%	-	-	-	-	-
Net income	31.0%	0.3%	14.0%	-5.8%	51.1%	11.7%	15.7%	9.3%	19.2%	36.5%	21.3%	16.0%	-	-	-	-	-
YoY growth (%)																	
Revenue	68.0%	45.7%	44.0%	34.3%	48.7%	57.7%	57.2%	66.7%	51.0%	77.4%	92.0%	105.7%	56%	46%	58%	83%	49%
Gross profit	76.3%	61.3%	60.7%	34.9%	46.3%	66.2%	70.2%	95.3%	70.8%	91.1%	103.3%	114.7%	59%	56%	70%	97%	53%
Operating income	79.0%	58.8%	57.6%	40.0%	55.5%	81.1%	95.0%	115.5%	79.5%	105.7%	115.5%	128.7%	65%	57%	88%	110%	58%
Net income	75.3%	42.8%	57.6%	41.1%	62.8%	81.3%	84.0%	113.4%	68.3%	105.7%	115.5%	128.7%	75%	53%	86%	107%	58%

Source: Company data, Goldman Sachs Global Investment Research

TUC

We revise up TUC's 12m TP from NT$900 to NT$1,015, to factor in our more positive view on the target P/E multiple (from 20x 4Q26-3Q27E EPS previously to 22x 4Q26-3Q27E EPS now, which is in-line with the AI PCB industry average valuation in the past 3 years), to better reflect the solid AI demand outlook into the long term and the much more favorable than expected pricing outlook in coming quarters/years, and factor in our earnings revision.

Exhibit 6: TUC P&L table

NT$mn	1Q25	2Q25	3Q25	4Q25	1Q26E	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025E	2026E	2027E	2028E
Revenue	6,372	6,780	8,063	9,125	10,165	12,515	14,442	14,620	15,552	18,197	24,112	27,147	23,070	30,340	51,741	85,009	130,739
Gross profit	1,533	1,435	1,915	2,015	2,409	3,277	4,200	4,407	4,781	5,507	7,474	8,601	5,342	6,898	14,293	26,364	42,408
Operating expense	(597)	(583)	(649)	(725)	(773)	(876)	(953)	(994)	(1,026)	(1,128)	(1,302)	(1,412)	(2,010)	(2,554)	(3,596)	(4,868)	(7,005)
Operating income	936	853	1,266	1,290	1,636	2,401	3,247	3,412	3,755	4,379	6,172	7,190	3,332	4,344	10,697	21,495	35,403
Pretax income	935	840	1,318	1,426	1,600	2,381	3,247	3,412	3,755	4,379	6,172	7,190	3,378	4,519	10,641	21,495	35,332
Taxes expense	(263)	(188)	(315)	(344)	(352)	(524)	(779)	(819)	(939)	(1,095)	(1,543)	(1,797)	(773)	(1,109)	(2,474)	(5,374)	(8,833)
Net income	672	652	1,003	1,082	1,248	1,857	2,468	2,593	2,816	3,284	4,629	5,392	2,604	3,409	8,167	16,121	26,499
EPS, NT$	**2.43**	**2.36**	**3.58**	**3.76**	**4.33**	**6.45**	**8.57**	**9.00**	**9.78**	**11.40**	**16.07**	**18.72**	**9.56**	**12.13**	**28.35**	**55.96**	**91.98**
Ratio analysis and assumption																	
As % of sales																	
Gross margin	24.1%	21.2%	23.7%	22.1%	23.7%	26.2%	29.1%	30.1%	30.7%	30.3%	31.0%	31.7%	23.2%	22.7%	27.6%	31.0%	32.4%
Operating expense ratio	9.4%	8.6%	8.1%	7.9%	7.6%	7.0%	6.6%	6.8%	6.6%	6.2%	5.4%	5.2%	8.7%	8.4%	6.9%	5.7%	5.4%
Operating margin	14.7%	12.6%	15.7%	14.1%	16.1%	19.2%	22.5%	23.3%	24.1%	24.1%	25.6%	26.5%	14.4%	14.3%	20.7%	25.3%	27.1%
Net margin	10.6%	9.6%	12.4%	11.9%	12.3%	14.8%	17.1%	17.7%	18.1%	18.0%	19.2%	19.9%	11.3%	11.2%	15.8%	19.0%	20.3%
QoQ growth (%)																	
Revenue	1.0%	6.4%	18.9%	13.2%	11.4%	23.1%	15.4%	1.2%	6.4%	17.0%	32.5%	12.6%	-	-	-	-	-
Gross profit	4.7%	-6.4%	33.4%	5.2%	19.6%	36.0%	28.2%	4.9%	8.5%	15.2%	35.7%	15.1%	-	-	-	-	-
Operating income	0.4%	-8.9%	48.4%	1.9%	26.9%	46.7%	35.2%	5.1%	10.0%	16.6%	40.9%	16.5%	-	-	-	-	-
Net income	-4.6%	-3.1%	53.8%	8.0%	15.3%	48.8%	32.9%	5.1%	8.6%	16.6%	40.9%	16.5%	-	-	-	-	-
YoY growth (%)																	
Revenue	43.7%	18.9%	21.8%	44.6%	59.5%	84.6%	79.1%	60.2%	53.0%	45.4%	67.0%	85.7%	44%	32%	71%	64%	54%
Gross profit	51.5%	5.3%	27.4%	37.7%	57.1%	128.3%	119.4%	118.7%	98.5%	68.1%	77.9%	95.2%	69%	29%	107%	84%	61%
Operating income	65.8%	-1.6%	30.6%	38.4%	74.8%	181.6%	156.6%	164.6%	129.5%	82.4%	90.1%	110.7%	134%	30%	146%	101%	65%
Net income	48.8%	-6.0%	33.0%	53.6%	85.6%	185.0%	146.2%	139.6%	125.6%	76.8%	87.6%	107.9%	216%	31%	140%	97%	64%

Source: Company data, Goldman Sachs Global Investment Research

GCE

We revise up our GCE 12m TP from NT$1,060 to NT$1,210, with the same target PE multiple (22x; in-line with the AI PCB industry average valuation in the past 3 years), but rollover our valuation period to 4Q26-3Q27E (from 2H26-1H27) to better factor in the longer term positive earnings outlook, and factor in our earnings revision.

Exhibit 7: GCE P&L table

NT$mn	1Q25	2Q25	3Q25	4Q25	1Q26E	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026E	2027E	2028E
Revenue	12,063	13,853	17,680	16,408	18,492	22,912	27,879	31,650	29,995	36,719	44,292	48,190	38,952	60,004	100,933	159,195	191,681
Gross profit	3,776	4,095	6,288	5,524	6,343	8,411	10,529	11,800	11,210	14,430	17,681	19,393	11,396	19,683	37,082	62,715	77,001
Operating expense	(1,233)	(1,228)	(1,641)	(1,538)	(1,664)	(1,970)	(2,286)	(2,500)	(2,551)	(3,011)	(3,543)	(3,711)	(3,329)	(5,640)	(8,421)	(12,816)	(15,330)
Operating income	2,543	2,868	4,647	3,986	4,679	6,440	8,243	9,299	8,660	11,419	14,138	15,682	8,067	14,043	28,661	49,899	61,671
Pretax income	2,653	2,509	4,730	4,281	4,679	6,440	8,743	8,299	7,994	10,753	13,472	16,682	8,490	14,173	28,161	48,901	61,221
Taxes expense	(900)	(815)	(1,501)	(1,351)	(1,404)	(2,061)	(2,798)	(2,656)	(2,558)	(3,441)	(4,042)	(4,671)	(2,875)	(4,567)	(8,918)	(14,712)	(18,604)
Net income	1,753	1,694	3,229	2,930	3,275	4,380	5,945	5,643	5,436	7,312	9,430	12,011	5,616	9,607	19,243	34,190	42,618
EPS, NT$	**3.60**	**3.48**	**6.53**	**5.86**	**6.55**	**8.76**	**11.89**	**11.29**	**10.87**	**14.62**	**18.86**	**24.02**	**11.54**	**19.47**	**38.49**	**68.38**	**85.24**
Ratio analysis and assumption																	
As % of sales																	
Gross margin	31.3%	29.6%	35.6%	33.7%	34.3%	36.7%	37.8%	37.3%	37.4%	39.3%	39.9%	40.2%	29.3%	32.8%	36.7%	39.4%	40.2%
Operating expense ratio	10.2%	8.9%	9.3%	9.4%	9.0%	8.6%	8.2%	7.9%	8.5%	8.2%	8.0%	7.7%	8.5%	9.4%	8.3%	8.1%	8.0%
Operating margin	21.1%	20.7%	26.3%	24.3%	25.3%	28.1%	29.6%	29.4%	28.9%	31.1%	31.9%	32.5%	20.7%	23.4%	28.4%	31.3%	32.2%
Net margin	14.5%	12.2%	18.3%	17.9%	17.7%	19.1%	21.3%	17.8%	18.1%	19.9%	21.3%	24.9%	14.4%	16.0%	19.1%	21.5%	22.2%
QoQ growth (%)																	
Revenue	22.4%	14.8%	27.6%	-7.2%	12.7%	23.9%	21.7%	13.5%	-5.2%	22.4%	20.6%	8.8%	-	-	-	-	-
Gross profit	47.7%	8.5%	53.5%	-12.1%	14.8%	32.6%	25.2%	12.1%	-5.0%	28.7%	22.5%	9.7%	-	-	-	-	-
Operating income	55.5%	12.8%	62.0%	-14.2%	17.4%	37.6%	28.0%	12.8%	-6.9%	31.9%	23.8%	10.9%	-	-	-	-	-
Net income	36.7%	-3.4%	90.6%	-9.3%	11.8%	33.7%	35.7%	-5.1%	-3.7%	34.5%	29.0%	27.4%	-	-	-	-	-
YoY growth (%)																	
Revenue	33.1%	44.7%	69.1%	66.5%	53.3%	65.4%	57.7%	92.9%	62.2%	60.3%	58.9%	52.3%	30%	54%	68%	58%	20%
Gross profit	55.0%	35.3%	86.2%	116.1%	68.0%	105.4%	67.5%	113.6%	76.7%	71.6%	67.9%	64.4%	48%	73%	88%	69%	23%
Operating income	47.3%	30.3%	85.5%	143.7%	84.0%	124.6%	77.4%	133.3%	85.1%	77.3%	71.5%	68.6%	57%	74%	104%	74%	24%
Net income	44.0%	11.4%	102.5%	128.5%	86.8%	158.5%	84.1%	92.6%	66.0%	67.0%	58.6%	112.8%	59%	71%	100%	78%	25%

Source: Company data, Goldman Sachs Global Investment Research

ZDT

We revise up our ZDT 12m TP from NT$228 to NT$260, with the same target SOTP based valuation methodology (1) 2.6x blended P/B for ABF & BT substrate (in-line with early upcycle industry average), (2) 20x P/E for high-growth AI server/switch related PCB (in line with industry average), and (3) 8x P/E for low-growth consumer electronics PCB (in-line with industry downcycle average valuation), but rollover our valuation period from 2H26-1H27 to 4Q26-1Q27, and factor in our earnings revision.

Exhibit 8: ZDT P&L table

NT$mn	1Q25	2Q25	3Q25	4Q25	1Q26E	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025E	2026E	2027E	2028E
Revenue	40,082	38,203	47,366	56,870	43,691	50,466	60,788	66,296	56,907	61,670	78,811	90,288	171,664	182,522	221,241	287,676	362,926
Gross profit	5,885	7,011	10,410	12,830	6,606	9,540	14,567	16,265	11,062	13,435	20,685	23,474	32,461	36,136	46,979	68,657	91,187
Operating expense	(4,829)	(4,586)	(5,880)	(6,909)	(5,724)	(6,914)	(6,504)	(6,828)	(6,260)	(7,215)	(8,275)	(8,668)	(20,875)	(22,205)	(25,970)	(30,418)	(37,943)
Operating income	1,056	2,425	4,530	5,921	883	2,626	8,063	9,437	4,802	6,220	12,410	14,807	11,586	13,932	21,009	38,239	53,244
Pretax income	1,457	2,272	4,661	5,673	983	2,726	8,163	10,437	4,902	6,320	12,510	15,807	15,045	14,063	22,309	39,539	54,544
Taxes expense	(431)	(885)	(1,072)	(1,070)	(167)	(463)	(1,224)	(1,566)	(833)	(1,074)	(1,876)	(2,371)	(1,948)	(3,458)	(3,420)	(6,155)	(8,523)
Net income	632	605	2,392	3,161	585	1,603	4,898	6,262	2,917	3,716	7,506	9,484	9,180	6,791	13,348	23,623	32,575
EPS, NT$	**0.66**	**0.63**	**2.46**	**3.16**	**0.58**	**1.60**	**4.90**	**6.26**	**2.92**	**3.71**	**7.50**	**9.48**	**9.67**	**6.91**	**13.34**	**23.61**	**32.56**
Ratio analysis and assumption																	
As % of sales																	
Gross margin	14.7%	18.4%	22.0%	22.6%	15.1%	18.9%	24.0%	24.5%	19.4%	21.8%	26.2%	26.0%	18.9%	19.8%	21.2%	23.9%	25.1%
Operating expense ratio	12.0%	12.0%	12.4%	12.1%	13.1%	13.7%	10.7%	10.3%	11.0%	11.7%	10.5%	9.6%	12.2%	12.2%	11.7%	10.6%	10.5%
Operating margin	2.6%	6.3%	9.6%	10.4%	2.0%	5.2%	13.3%	14.2%	8.4%	10.1%	15.7%	16.4%	6.7%	7.6%	9.5%	13.3%	14.7%
Net margin	1.6%	1.6%	5.0%	5.6%	1.3%	3.2%	8.1%	9.4%	5.1%	6.0%	9.5%	10.5%	5.3%	3.7%	6.0%	8.2%	9.0%
QoQ growth (%)																	
Revenue	-28.6%	-4.7%	24.0%	20.1%	-23.2%	15.5%	20.5%	9.1%	-14.2%	8.4%	27.8%	14.6%	-	-	-	-	-
Gross profit	-48.7%	19.1%	48.5%	23.2%	-48.5%	44.4%	52.7%	11.7%	-32.0%	21.5%	54.0%	13.5%	-	-	-	-	-
Operating income	-81.0%	129.6%	86.8%	30.7%	-85.1%	197.4%	207.1%	17.0%	-49.1%	29.5%	99.5%	19.3%	-	-	-	-	-
Net income	-85.5%	-4.3%	295.3%	32.2%	-81.5%	174.1%	205.6%	27.9%	-53.4%	27.4%	102.0%	26.4%	-	-	-	-	-
YoY growth (%)																	
Revenue	23.3%	17.9%	-6.4%	1.3%	9.0%	32.1%	28.3%	16.6%	30.2%	22.2%	29.6%	36.2%	13%	6%	21%	30%	26%
Gross profit	10.3%	65.1%	-8.7%	11.9%	12.3%	36.1%	39.9%	26.8%	67.4%	40.8%	42.0%	44.3%	18%	11%	30%	46%	33%
Operating income	42.2%	-476.8%	-23.6%	6.6%	-16.4%	8.3%	78.0%	59.4%	443.9%	136.9%	53.9%	56.9%	26%	20%	51%	82%	39%
Net income	-35.3%	24.5%	-28.7%	-27.5%	-7.5%	164.9%	104.8%	98.1%	398.7%	131.9%	53.2%	51.5%	48%	-26%	97%	77%	38%

Source: Company data, Goldman Sachs Global Investment Research

Investment thesis, PT methodology and risks

EMC, a key supplier of high-end HDI material (70%+ market share) and SLP material (90%+ market share), has been focusing on the high-speed switch/server CCL market for 3+ years. We are positive on its market share trajectory in the server industry and expect EMC to increase its share from <10% in the Purley generation to 15-20%/20+% in the Whitley/Eagle Stream platforms. EMC is also proactively expanding its share in the switch market (40%/30%+ in 2024/23 vs. <5% before 2019), with an expanding AI customer base (only one 400G customer in 3Q20, to being one of the largest suppliers in 2023), which should continue to benefit the company's top/bottom line growth over time. We believe EMC's leading position in the AI server CCL market should drive strong revenue growth and GM/OPM expansion in the long term, given that it is the major supplier for all Nvidia/Google/AWS AI server projects, and we expect it to maintain its leadership position. Considering its leading position in the high-end CCL industry, solid production skills and earnings growth outlook, we are Buy rated on the stock. With the shares trading at a 2026E P/E below AI server component suppliers and supported by a solid growth outlook, we view valuation as attractive.

Key downside risks include: (1) RCC to replace all high-end smartphone HDI design; (2) rising trade tensions, which could lead to weaker smartphone and server shipments; and (3) rising competition from mainland China peers

Valuation methodology: Our 12m TP of NT$3,500 is based on 27x 4Q26E-3Q27E P/E (in line with the peak P/E multiple for AI component suppliers in past 3 years).

Key downside risks: (1) RCC to replace all high-end smartphone HDI design; (2) rising trade tensions, which could lead to weaker smartphone and server shipments; and (3) rising competition from mainland China peers.

TUC is a key high-end global CCL supplier that focuses on M7+ grade high-speed CCL (key applications include high-end switches (100G+) and AI server materials), with 20%+ market share in the past two years. The company continues to expect its unit market share in the server industry to increase from 15% in Whitley and Purley to higher in the Eagle Stream generation, given its plans to launch two types of products for low-end and high-end customers (T2A and T2C) to gain market share. Also, strong growth in 400G/800G switch shipments and new 800G switch shipments, scheduled for launch in 2025-27, should continue to be a key demand driver for TUC, in our view. Additionally, the company is working on high-end AI projects with CSP and enterprise players, which we believe should drive revenue momentum in the long term. We view the stock as undervalued vs. Taiwan CCL peers on a P/E basis and rate it Buy.

Key downside risks include: (1) slower-than-expected share gains in the low-loss CCL segment; (2) rising trade tensions, which could lead to weaker server and switch shipments globally; and (3) rising competition from mainland China peers.

Valuation methodology: Our 12m TP of NT$1,015 is based on 22x 4Q26-3Q27 P/E (+2x STDV higher than the past 3-year industry average P/E multiple).

Key downside risks: (1) slower-than-expected share gains in the low-loss CCL segment; (2) rising trade tensions, which could lead to weaker server and switch shipments globally; and (3) rising competition from mainland China peers.

GCE is a key cloud (server+switch) PCB supplier globally with 20%+ market share in 2022. We expect cloud PCB market demand to grow at a 15% 2024E-26E CAGR (far stronger than the overall RPCB+HDI market demand growth rate at 5% 2024E-26E CAGR). We believe GCE's proactive capacity expansion plan (33% in 2026) and high annualized ROI (300%+) will continue to drive the company's revenue/margin level in the coming quarters/years. Additionally, we believe GCE can continue to see strong demand from the increasing content per server/switch, as we expect both markets to see a 20-50%+ layer count increase after the generation upgrade (upgrade to 400G/Eagle Stream). We are constructive on GCE's long-term growth opportunity given its solid position in the rapidly growing cloud PCB market, and we are Buy rated on the stock. GCE is trading higher than its historical P/E average level driven by strong AI server demand, but lower than the TW AI server players' average.

Price Target Risks and Methodology

Valuation methodology: Our 12m TP of NT$1,210 is based on a 22x 4Q26-3Q27E P/E (in-line with the AI PCB leader valuation in past 3 years).

Key downside risks: (1) weaker-than-expected AI server market demand; (2) delays in the company's Nvidia OAM/UBB product qualification, (3) weaker-than-expected HDI demand, and (4) high than expected ramp up cost for new capacity

Zhen Ding Technology (ZDT) is the global leading PCB supplier (top 1 in market share in sales) with 7-10% market share, and started its ABF substrate business from 2023 in Shenzhen, China, and has the highest ABF revenue exposure to the China market compared to all other ABF substrate suppliers. We believe ZDT will hold more than 40% ABF substrate market share in China by 2027E (up from 31% in 2024) and benefit the most from the strong and growing China ABF substrate market (53% 2025E-27E CAGR) due to the increasing in-sourcing ratio for the AI computing ASIC/GPU market in China. We also expect the increasing foldable smartphone penetration rate in the long term to drive ZDT's FPC TAM, and account for 55%+ of the company's total revenue in 2027E. We see further revenue upside potential from foldable phones. We are Buy rated mainly due to our positive view on the company's China ABF substrate business opportunity, which should not only drive ZDT's revenue growth, but also improve its OPM level in the long term. We see upside for the stock price considering the company's solid growth opportunity, and our TP implied P/E multiple is still lower than the company's past 10+ year P/E average, which we believe is fair considering the potential upside from China ABF substrate market and increasing foldable phone penetration rate.

Valuation: Our 12-month TP of NT$260 is based on a SOTP valuation methodology, including (1) 2.6x blended 4Q26-3Q27E P/B for ABF & BT substrate, (2) 20x 4Q26-3Q27E P/E for high-growth AI server/switch related PCB (in line with industry average), and (3) 8x 4Q26-3Q27E P/E for low-growth consumer electronics PCB (in-line with industry downcycle average valuation).

Key downside risks include: (1) weaker-than-expected iPhone demand or slower content upgrade, (2) delay in new substrate capacity or slower-than-expected yield rate ramp up progress, and (3) fiercer competition in China ABF market.

Disclosure Appendix

Reg AC

I, Chao Wang, hereby certify that all of the views expressed in this report accurately reflect my personal views about the subject company or companies and its or their securities. I also certify that no part of my compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

Rating and pricing information

MediaTek (Buy, NT$1,680.00)

The rating(s) for Elite Material, GCE, Taiwan Union Technology Corp. and Zhen Ding Technology Holding is/are relative to the other companies in its/their coverage universe: AWSC, Airtac International Group, Delta Electronics, Elite Material, GCE, Hiwin Corp., ITEQ Corp, Kinsus, Lotes, NYPCB, Taiwan Union Technology Corp., Unimicron Technology, Win Semiconductors Corp., Yageo Corp., Zhen Ding Technology Holding

Company-specific regulatory disclosures

The following disclosures relate to relationships between The Goldman Sachs Group, Inc. (with its affiliates, "Goldman Sachs") and companies covered by Goldman Sachs Global Investment Research and referred to in this research.

Goldman Sachs beneficially owned 1% or more of common equity (excluding positions managed by affiliates and business units not required to be aggregated under US securities law) as of the month end preceding this report: Elite Material (NT$2,810.00), Taiwan Union Technology Corp. (NT$554.00) and Zhen Ding Technology Holding (NT$209.50)

Goldman Sachs expects to receive or intends to seek compensation for investment banking services in the next 3 months: Elite Material (NT$2,810.00) and Zhen Ding Technology Holding (NT$209.50)

Goldman Sachs had an investment banking services client relationship during the past 12 months with: Elite Material (NT$2,810.00) and Zhen Ding Technology Holding (NT$209.50)

There are no company-specific disclosures for: GCE (NT$988.00)

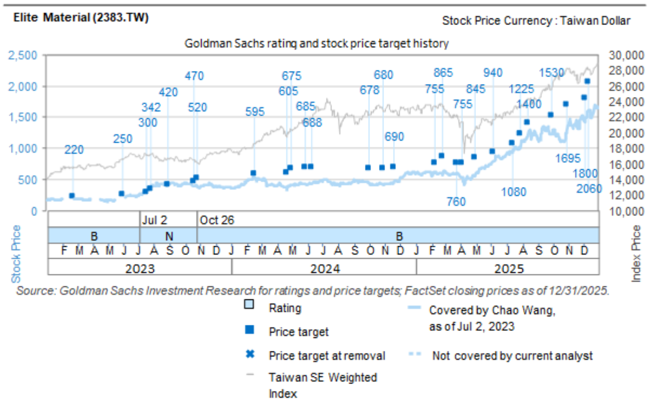
Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

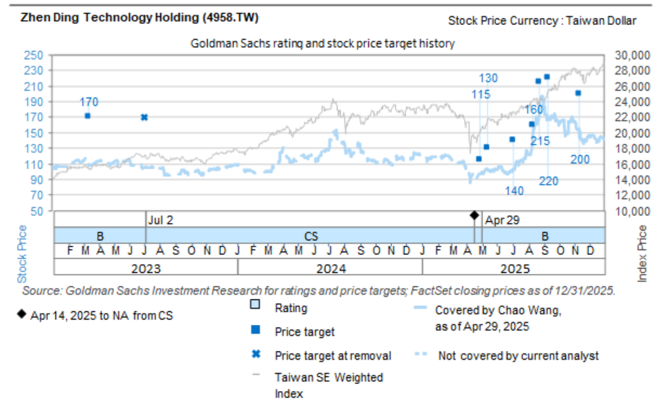
	Rating Distribution			Investment Banking Relationships		
	Buy	Hold	Sell	Buy	Hold	Sell
Global	50%	34%	16%	65%	61%	46%

As of January 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,055 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

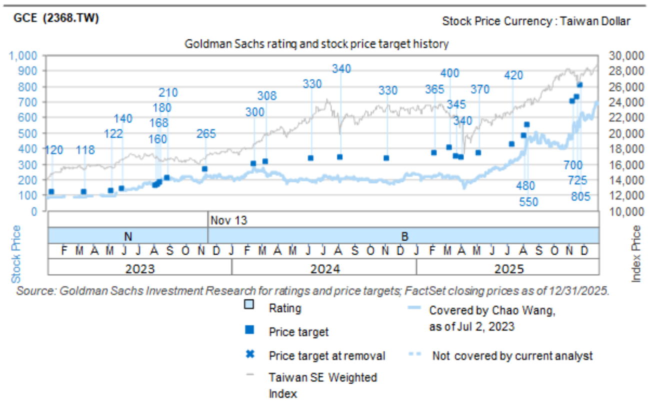
Price target and rating history chart(s)



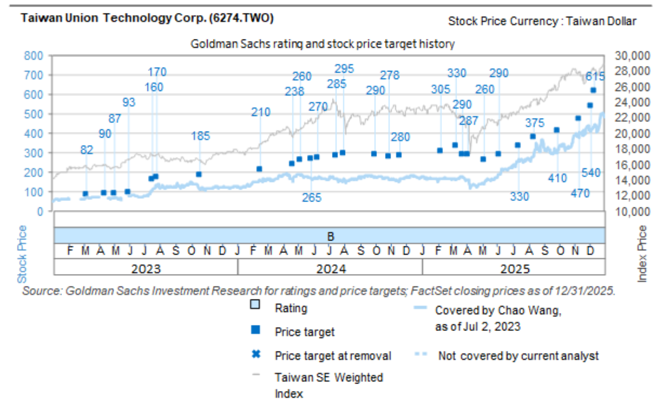
The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.

Target price history table(s)

Taiwan Union Technology Corp. (6274.TWO)

Date of report	Target price (NT$)	Closing price (NT$)
12-Mar-26	900.00	452.50
05-Feb-26	740.00	520.00
06-Jan-26	650.00	476.00
12-Dec-25	615.00	443.00
04-Dec-25	540.00	411.00
12-Nov-25	470.00	407.00
29-Sep-25	410.00	298.00
13-Aug-25	375.00	282.50
15-Jul-25	330.00	264.00
05-Jun-25	290.00	188.50
07-May-25	260.00	142.50
06-Apr-25	287.00	159.00
24-Mar-25	290.00	176.50
12-Mar-25	330.00	160.50
09-Feb-25	305.00	163.50
20-Nov-24	280.00	155.00
30-Oct-24	278.00	162.00
03-Oct-24	290.00	175.00
31-Jul-24	295.00	160.00
17-Jul-24	285.00	168.00
11-Jun-24	270.00	164.50
29-May-24	265.00	167.00
08-May-24	260.00	185.50

GCE (2368.TW)

Date of report	Target price (NT$)	Closing price (NT$)
10-Mar-26	1,060.00	817.00
06-Jan-26	925.00	662.00
26-Nov-25	805.00	610.00
19-Nov-25	725.00	522.00
11-Nov-25	700.00	498.50
13-Aug-25	550.00	460.00
07-Aug-25	480.00	372.00
15-Jul-25	420.00	320.00
09-May-25	370.00	226.00
06-Apr-25	340.00	195.00
24-Mar-25	345.00	214.00
11-Mar-25	400.00	209.00
09-Feb-25	365.00	228.00
07-Nov-24	330.00	188.50
07-Aug-24	340.00	207.00
11-Jun-24	330.00	187.50
12-Mar-24	308.00	242.50
18-Feb-24	300.00	275.00
13-Nov-23	265.00	217.00
30-Aug-23	210.00	208.50
16-Aug-23	180.00	188.00
10-Aug-23	168.00	166.50
07-Aug-23	160.00	168.00

Taiwan Union Technology Corp. (6274.TWO)

Date of report	Target price (NT$)	Closing price (NT$)
23-Apr-24	238.00	164.50
18-Feb-24	210.00	156.50
20-Oct-23	185.00	111.00
27-Jul-23	170.00	133.50
19-Jul-23	160.00	108.00
01-Jun-23	93.00	74.10
04-May-23	87.00	65.00
14-Apr-23	90.00	69.20

GCE (2368.TW)

Date of report	Target price (NT$)	Closing price (NT$)
01-Jun-23	140.00	125.00
11-May-23	122.00	95.00

Elite Material (2383.TW)

Date of report	Target price (NT$)	Closing price (NT$)
12-Mar-26	3,200.00	2,600.00
05-Feb-26	2,585.00	1,950.00
06-Jan-26	2,250.00	1,695.00
12-Dec-25	2,060.00	1,605.00
04-Dec-25	1,800.00	1,425.00
30-Oct-25	1,695.00	1,275.00
29-Sep-25	1,530.00	1,225.00
13-Aug-25	1,400.00	1,200.00
30-Jul-25	1,225.00	1,005.00
15-Jul-25	1,080.00	970.00
05-Jun-25	940.00	789.00
30-Apr-25	845.00	556.00
06-Apr-25	755.00	527.00
24-Mar-25	760.00	604.00
25-Feb-25	865.00	576.00
09-Feb-25	755.00	615.00
20-Nov-24	690.00	447.50
30-Oct-24	680.00	419.50
03-Oct-24	678.00	442.00
11-Jun-24	688.00	422.50
29-May-24	685.00	427.00
30-Apr-24	675.00	412.00
23-Apr-24	605.00	376.00
18-Feb-24	595.00	534.00
26-Oct-23	520.00	388.50
20-Oct-23	470.00	403.00
30-Aug-23	420.00	415.50
26-Jul-23	342.00	354.00
19-Jul-23	300.00	318.00
01-Jun-23	250.00	210.00

Zhen Ding Technology Holding (4958.TW)

Date of report	Target price (NT$)	Closing price (NT$)
13-Mar-26	228.00	188.50
12-Nov-25	200.00	144.50
10-Sep-25	220.00	168.50
25-Aug-25	215.00	180.50
12-Aug-25	160.00	147.00
04-Jul-25	140.00	103.00
13-May-25	130.00	104.50
29-Apr-25	115.00	97.10

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